

A. An illustration of error propagation

Illustrative Example. Consider the following matrix multiplication sequence involving two consecutive linear layers:

$$\underbrace{\begin{bmatrix} 3 & 6 \end{bmatrix}}_X \cdot \underbrace{\begin{bmatrix} 2 & 2 \\ 4 & 1 \end{bmatrix}}_{W_1} \cdot \underbrace{\begin{bmatrix} 4 & 4 \\ 8 & 1 \end{bmatrix}}_{W_2} = \begin{bmatrix} 192 & 115 \end{bmatrix}.$$

The *local loss* is defined as $\|X \cdot (M \odot W_1 - W_1)\|_F^2$, and the *downstream loss* as $\|X \cdot (M \odot W_1 - W_1) \cdot W_2\|_F^2$, where M is the binary mask. Now consider pruning two different weights in W_1 .

Pruning entry $W_{1,1} = 2$ (Mask 1) results in a local loss of 36 and a downstream loss of 1152. In contrast, pruning entry $W_{2,2} = 1$ (Mask 2) yields the same local loss of 36, but a significantly larger downstream loss of 2340. This toy example illustrates that pruning decisions which are indistinguishable under local error metrics may lead to dramatically different downstream errors due to *error propagation*. As layer depth increases, this effect compounds, motivating the need for pruning strategies that can *foresee* such interactions.

B. ForeSight for Attention and MLP

ForeSight extends to self-attention by pruning the *query* (Q) and *key* (K) projections *jointly*. Throughout, let $X \in \mathbb{R}^{l \times d}$ denote the flattened input activations of a calibration batch ($l = b \times s$, where b is the batch size and s the per-sample sequence length).

Weights and adapters.

- Base projections: $W^Q, W^K \in \mathbb{R}^{d \times d_k}$ and $W^V \in \mathbb{R}^{d \times d_v}$.
- LoRA adapters: $B^Q, B^K \in \mathbb{R}^{d \times r}$, $A^Q, A^K \in \mathbb{R}^{r \times d_k}$ and B^V, A^V for the value path.

The effective projections are $\widetilde{W}^Q = W^Q + B^Q A^Q$ and $\widetilde{W}^K = W^K + B^K A^K$.

Two-layer interaction. The attention “kernel” observed by the softmax is

$$X \widetilde{W}^Q \widetilde{W}^{K\top} X^\top.$$

Masking the Q projection. With a binary mask $M^Q \in \{0, 1\}^{d \times d_k}$ and sparsity constraint $\|M^Q\|_0 \leq k$, ForeSight selects the k coordinates whose removal minimizes the forward error

$$\min_{M^Q} \|X [\widetilde{W}^Q - M^Q \odot \widetilde{W}^Q] \widetilde{W}^{K\top}\|_F^2.$$

A second-order approximation yields the importance score

$$\Delta \mathcal{L}_{ij} \propto |\widetilde{W}_{ij}^Q| \|\widetilde{W}_{j,:}^K\|_2 \|X_{:,i}\|_2.$$

Masking the K projection. An analogous problem is solved for $M^K \in \{0, 1\}^{d \times d_k}$:

$$\min_{M^K} \|\widetilde{W}^Q [\widetilde{W}^K - M^K \odot \widetilde{W}^K]^\top X^\top\|_F^2, \quad \|M^K\|_0 \leq k,$$

with score

$$\Delta \mathcal{L}_{ij} \propto |\widetilde{W}_{ij}^K| \|\widetilde{W}_{:,j}^Q\|_2 \|X_{:,i}\|_2.$$

These propagation-aware scores prune Q and K in one pass, ensuring that early-layer sparsity respects downstream attention dynamics.

Masking the output projection O . The attention output is projected back to the residual stream by $\widetilde{W}^O = W^O + B^O A^O \in \mathbb{R}^{d \times d}$. Its output is consumed by the MLP *input* of both the gate and up projections. We therefore weight each output coordinate by the average downstream sensitivity

$$\nu_u^O := \frac{1}{2} \left(\|\widetilde{W}_{:,u}^{\text{gate}}\|_2 + \|\widetilde{W}_{:,u}^{\text{up}}\|_2 \right), \quad u \in [d],$$

and score each entry of $\widetilde{W}^O$ by

$$\Delta \mathcal{L}_{uv}^O \propto |\widetilde{W}_{uv}^O| \|X_{:,v}\|_2 \nu_u^O.$$

This matches the implementation $|\widetilde{W}^O| \odot X_{\text{hat}} \odot \nu^O$ (with broadcasting).

Masking MLP gate/up projections. For the gated MLP, let $\widetilde{W}^{\text{gate}}, \widetilde{W}^{\text{up}} \in \mathbb{R}^{d_{ff} \times d}$ and $\widetilde{W}^{\text{down}} \in \mathbb{R}^{d \times d_{ff}}$ denote the effective (projection + LoRA) weights. Both gate and up outputs are mapped back by $\widetilde{W}^{\text{down}}$, so we use the per-hidden-unit propagation norm

$$\nu_h^{\text{down}} := \|\widetilde{W}_{:,h}^{\text{down}}\|_2, \quad h \in [d_{ff}].$$

Ignoring elementwise nonlinearities (Appendix C), ForeSight assigns the importance scores

$$\Delta \mathcal{L}_{hi}^{\text{gate}} \propto |\widetilde{W}_{hi}^{\text{gate}}| \|X_{:,i}\|_2 \nu_h^{\text{down}}, \quad \Delta \mathcal{L}_{hi}^{\text{up}} \propto |\widetilde{W}_{hi}^{\text{up}}| \|X_{:,i}\|_2 \nu_h^{\text{down}}.$$

C. Derivation of ForeSight importance Score

Let $\mathbf{X} \in \mathbb{R}^{m \times d}$ be a fixed data matrix and let $\mathbf{U}_1 = \mathbf{W}_1 + \mathbf{B}_1 \mathbf{A}_1 \in \mathbb{R}^{d \times h}$, $\mathbf{U}_2 = \mathbf{W}_2 + \mathbf{B}_2 \mathbf{A}_2 \in \mathbb{R}^{h \times p}$ be the (affine, bias-free) effective weight matrices of two consecutive linear layers. Define the loss

$$\mathcal{L}(\mathbf{U}_1) = \|\mathbf{X} \mathbf{U}_1 \mathbf{U}_2\|_F^2 = \text{Tr}(\mathbf{U}_2^\top \mathbf{U}_1^\top \mathbf{X}^\top \mathbf{X} \mathbf{U}_1 \mathbf{U}_2).$$

Fix an entry $\theta = \mathbf{U}_1[i, j]$. We study the loss change $\Delta \mathcal{L}_{ij}$ obtained by setting this entry to zero while *keeping all other parameters constant*. Throughout we assume (i) $\mathbf{X}$ and $\mathbf{U}_2$ are constant during the perturbation, (ii) $\mathbf{U}_1$ corresponds to a first-order stationary point of $\mathcal{L}$ (i.e. $\nabla_{\mathbf{U}_1} \mathcal{L} = \mathbf{0}$), and (iii) higher-than-second-order terms in a Taylor expansion are negligible for the magnitude of the perturbation considered.

Write $E_{ij} = e_i e_j^\top$, so that θE_{ij} extracts the affected entry. A standard matrix-calculus identity gives

$$\frac{\partial \mathcal{L}}{\partial \mathbf{U}_1} = 2 \mathbf{X}^\top \mathbf{X} \mathbf{U}_1 (\mathbf{U}_2 \mathbf{U}_2^\top), \quad (5)$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \theta} = 2 e_i^\top \mathbf{X}^\top \mathbf{X} \mathbf{U}_1 (\mathbf{U}_2 \mathbf{U}_2^\top) e_j. \quad (6)$$

Because $\mathbf{U}_1$ depends linearly on θ only through the i -th row, differentiation once more yields the diagonal Hessian entry

$$\frac{\partial^2 \mathcal{L}}{\partial \theta^2} = 2 e_i^\top \mathbf{X}^\top \mathbf{X} E_{ij} \mathbf{U}_2 \mathbf{U}_2^\top e_j = 2 (\mathbf{X}^\top \mathbf{X})_{ii} (\mathbf{U}_2 \mathbf{U}_2^\top)_{jj}.$$

Letting $\delta\theta = -\theta$ denote the pruning perturbation, a second-order Taylor expansion gives

$$\begin{aligned} \Delta \mathcal{L}_{ij} &= \mathcal{L}(\mathbf{U}_1 + \delta\theta E_{ij}) - \mathcal{L}(\mathbf{U}_1) \\ &= \underbrace{\frac{\partial \mathcal{L}}{\partial \theta}}_{=0} \delta\theta + \frac{1}{2} \frac{\partial^2 \mathcal{L}}{\partial \theta^2} (\delta\theta)^2 + O(\|\delta\theta\|^3) \\ &\approx \frac{1}{2} \theta^2 (\mathbf{X}^\top \mathbf{X})_{ii} (\mathbf{U}_2 \mathbf{U}_2^\top)_{jj}. \end{aligned} \quad (7)$$

Hence, under the stated assumptions, the *exact* second-order approximation to the loss increase is non-negative and is given by

$$\Delta\mathcal{L}_{ij} \approx \frac{1}{2} \theta_{ij}^2 (\mathbf{X}^\top \mathbf{X})_{ii} (\mathbf{U}_2 \mathbf{U}_2^\top)_{jj} \quad (i \in [d], j \in [h]).$$

This value serves as a principled importance score for weight pruning: the larger the product of the data-covariance entry $(\mathbf{X}^\top \mathbf{X})_{ii}$, the downstream energy $(\mathbf{U}_2 \mathbf{U}_2^\top)_{jj}$, and the squared weight magnitude θ_{ij}^2 , the greater the predicted loss incurred by zeroing θ_{ij} .

Nonlinearities in ForeSight can be safely ignored. In a preLayerNorm model, calibration pre-activations a is approximately standard normal, and the exact GELU has a slope in $[-0.129, 1.129]$. Thus, GELU only reweights token contributions by bounded factors, so our group selection by magnitudes is stable except for near ties. Empirically, the top k overlap between pre-activation and post-activation scoring is at least 90%.

D. Training Dataset Construction

The domain training corpora are listed in Tables 3 and 4. For each domain, we stratify the original training split by label, shuffle with a fixed seed (1234), and subsample to the desired size. In PUBMEDQA, we retain only the *yes* and *no* classes because the released training set contains no *maybe* examples. For all other datasets, labels with too few instances are upsampled by sampling with replacement until the target count is reached. All training data and evaluation scripts follow the Alpaca template (Dubois et al. 2023), with the full prompt formats listed in Appendix .

Table 3: **Health domain** training datasets with label distributions, shuffle settings, random seed, and sequence length.

Dataset	Label	Instances	Shuffle	Seed	Seq. Len.
MedNLI	entailment	2333	TRUE	1234	2048
	contradiction	2333		1234	2048
	neutral	2334		1234	2048
PubMedQA	yes	3500	TRUE	1234	2048
	no	3500		1234	2048
	maybe	0		1234	2048
HQS	–	1000	TRUE	1234	2048

Two domain-specific datasets are used in our experiments: **PubMedQA** and **CaseHold**. For **PubMedQA**, we sample 7,000 *yes* and 7,000 *no* examples; the *maybe* label is excluded due to lack of availability in the training split. For **CaseHold**, we sample 3,000 instances from each class label $\{0, 1, 2, 3, 4\}$. All dataset splits are shuffled once using a fixed seed

E. Hyperparameters Settings

We provide all the hyperparameter used for training in Table 1, Table 2, and the two task specific domain model used in Section . For all dense Health domain models, we use training dataset described in Table 3 and the evaluation datasets described in Table 9. For all dense Legal domain models, we use training dataset described in Table 4

Table 4: **Legal domain** training datasets with label distributions, shuffle settings, random seed, and sequence length.

Dataset	Label	Instances	Shuffle	Seed	Seq. Len.
ContractNLI	entailment	2100	TRUE	1234	2048
	contradiction	1200		1234	2048
	neutral	2700		1234	2048
CaseHold	0	1400	TRUE	1234	2048
	1	1400		1234	2048
	2	1400		1234	2048
	3	1400		1234	2048
	4	1400		1234	2048
BillSum	–	1000	TRUE	1234	2048

and the evaluation datasets described in Table 10. The training hyperparameter are set the be the same for PubMedQA model and CaseHold model with training datasets described in previous section. To note that all training are conducted on 4 NVIDIA A100 GPUs with 80GB memory. We adjust the per-device batch size and gradient accumulation steps to maintain a global batch size of 32.

Table 5: Dense Health Domain Training Hyperparameters

Hyperparameter	Value
model_name	meta-llama/Llama-2-7b-hf
learning_rate	1×10^{-4}
per_device_batch_size	2
gradient_accumulation_steps	4
num_train_epochs	3
weight_decay	0.01
fp16	True
save_steps	1000
max_seq_length	2048
lora_r	8
lora_alpha	16
lora_dropout	0
lora_bias	none
lora_target_modules	q_proj, o_proj, v_proj, k_proj, gate_proj, up_proj, down_proj
optimizer	torch.Adam

The LoRA training hyperparameter and training dataset used for EfficientXpert are identical to those used for the dense model to ensure a fair comparison. For ForeSight Mask, the mask learning rate is set to 0.5. For Partial Brain Surgeon, the regularization coefficient is set to 1×10^{-8} .

F. Evaluation Details

The name of the evaluation and number of instances in the test set are listed in Table 9 and Table 10. For all the evaluations, we adopt the Alpaca template (Dubois et al. 2023) to evaluate the performance of the models. All Alpaca templates are listed in Appendix .

G. Further Analysis

The Grassmann distance between two LoRA adapters of the same weight, parameterized by (B_1, A_1) and (B_2, A_2) re-

Table 6: Dense Health Domain Training Hyperparameters

Hyperparameter	Value
model_name	meta-llama/Llama-2-7b-hf
learning_rate	1×10^{-4}
per_device_batch_size	2
gradient_accumulation_steps	4
num_train_epochs	3
weight_decay	0.01
fp16	True
save_steps	1000
max_seq_length	2048
lora_r	8
lora_alpha	16
lora_dropout	0
lora_bias	none
lora_target_modules	q_proj, o_proj, v_proj, k_proj, gate_proj, up_proj, down_proj
optimizer	torch.Adam

Table 7: Dense Legal Domain Training Hyperparameters

Hyperparameter	Value
model_name	meta-llama/Llama-2-7b-hf
learning_rate	1×10^{-4}
per_device_batch_size	2
gradient_accumulation_steps	4
num_train_epochs	3
weight_decay	0.01
fp16	True
save_steps	1000
max_seq_length	2048
lora_r	8
lora_alpha	16
lora_dropout	0
lora_bias	none
lora_target_modules	q_proj, o_proj, v_proj, k_proj, gate_proj, up_proj, down_proj
optimizer	torch.Adam

spectively, is given by

$$d(U_1, U_2) = \left(\sum_{i=1}^r \cos^{-1}(\sigma_i)^2 \right)^{\frac{1}{2}}, \quad (8)$$

$$\text{where } B_1 A_1 = U_1 \Sigma_1 V_1^\top, \quad B_2 A_2 = U_2 \Sigma_2 V_2^\top. \quad (9)$$

and σ_i are the singular values of $U_1^\top U_2$, $r = 8$ for our experiment.

To understand how is task information encoded in the LoRA adapter, we further analyze the projection energy of each domain-specific or task-diverse model introduced in Section , as shown in Figure 4. For each model, we compute the projection energy of the LoRA adapters and average it by weight type. The plotted results reveal that the distribution of average projection energy exhibits **more consistent local patterns within the same task** (e.g., CaseHold vs. Legal or PubMedQA vs. Health), even when the domains differ.

For instance, **PubMedQA** and **Health**, although from different domains, both exhibit strong activation in

Table 8: Dense Legal Domain Training Hyperparameters

Hyperparameter	Value
model_name	meta-llama/Llama-3-8b
learning_rate	1×10^{-4}
per_device_batch_size	1
gradient_accumulation_steps	8
num_train_epochs	3
weight_decay	0.01
fp16	True
save_steps	1000
max_seq_length	2048
lora_r	8
lora_alpha	16
lora_dropout	0
lora_bias	none
lora_target_modules	q_proj, o_proj, v_proj, k_proj, gate_proj, up_proj, down_proj
optimizer	torch.Adam

Table 9: Overview of health datasets used for evaluation.

	InternalMed_Harrison	MedNLI	PubMedQA	HQS
Domain	Health	Health	Health	Health
Task / Type	Generation	NLI	QA	Summarization
# Instances in Test	300	1422	500	100
Metrics	Perplexity	Acc & Macro-F1	Acc & Macro-F1	ROUGE

gate_proj and up_proj. In contrast, **CaseHold** and **Legal** emphasize k_proj and up_proj. This pattern suggests that **task similarity plays a dominant role in shaping local projection energy patterns**, whereas **domain-level information may guide the broader global distribution**. These findings imply that LoRA adapters capture a **global structure governed by domain**, while their **local energy distribution reflects task-specific characteristics**, especially across components like attention, gate, up projection in MLP.

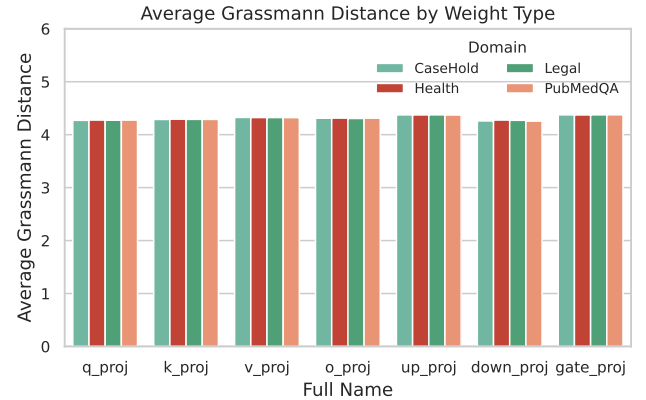


Figure 7: Grassmann distance between LoRA adapters and weight matrices by weight type.

These findings imply that while LoRA adapters may encode a global structure shaped by domain identity, task-specific characteristics drive significant variation in projection energy across weight types—particularly within attention and MLP components.

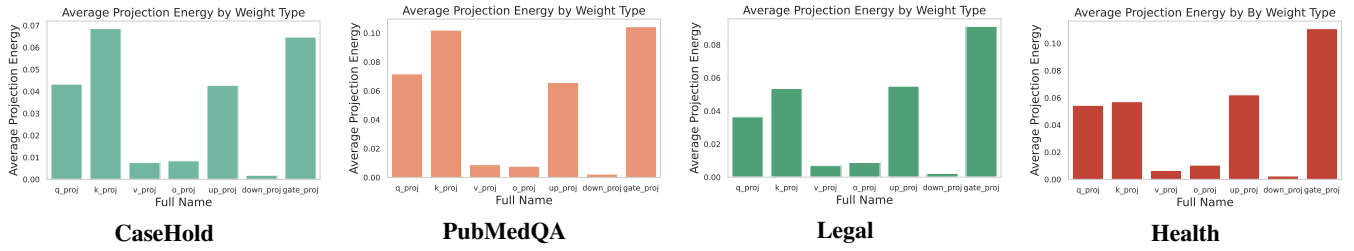


Figure 4: Projection energy by full name across domains.

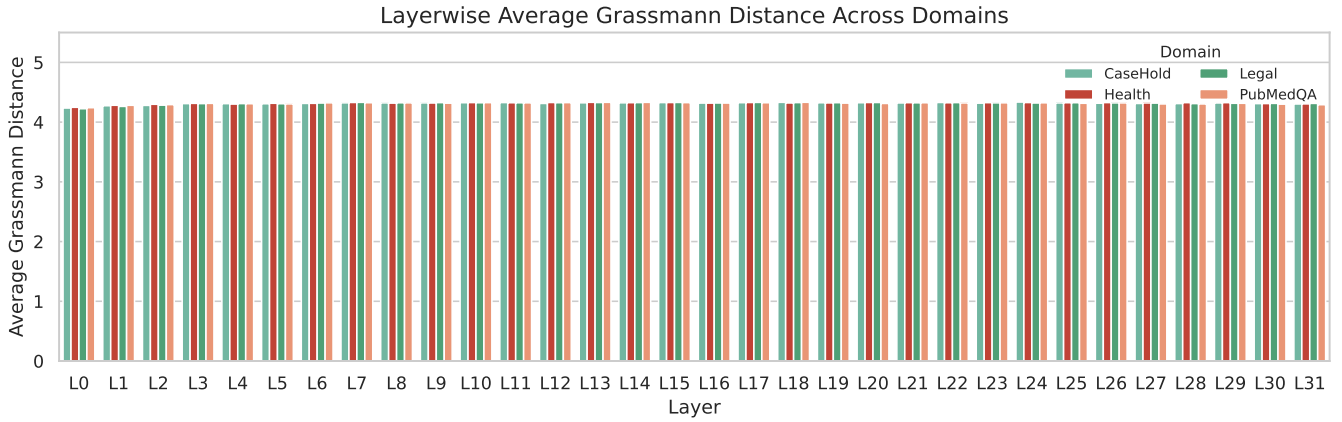


Figure 5: Grassmann distance between LoRA adapters and weight matrices across layers.

Table 10: Overview of legal datasets used for evaluation.

	MultiLegalFile	ContractNLI	CaseHOLD	BillSum
Domain	Legal	Legal	Legal	Legal
Task / Type	Generation	NLI	QA	Summarization
# Instances in Test	300	1991	200	200
Metrics	Perplexity	Acc & Macro-F1	Acc & Macro-F1	ROUGE

To further explore this contrast, we visualize the Grassmann distance between LoRA adapters and the corresponding base weight matrices for each layer. As shown in Figure 5, the Grassmann distances remain consistently high (above 4) across all layers and domains, indicating that LoRA adapters systematically amplify directions that diverge from the pretrained model’s dominant subspaces. This pattern is mirrored in Figure 7, where all weight types exhibit similarly large distances.

These results suggest a fundamental duality: while the global subspace shift induced by LoRA is governed by domain-level signals—producing consistently large Grassmann distances—the local alignment revealed by projection energy varies according to task-specific usage patterns.

Alpaca Templates

PubMedQA

1 "Below is an instruction that describes a task related to HealthCare, "

2 "paired with detailed input from a scientific article.\n\n"
3 "Instruction: As an expert doctor in clinical science and medical knowledge, "
4 "can you tell me if the following statement is correct? Answer yes, no, or maybe.\n\n"
5 "Input:\n"
6 "Contexts:\n{contexts}\n\n"
7 "Section Labels: {labels}\n\n"
8 "MeSH Terms: {meshes}\n\n"
9 "Question: {question}\n\n"
10 "Response: The answer is"

MedNLI

1 "Below is an instruction for analyzing clinical information.\n\n"
2 "Instruction:\n"
3 "You are a licensed physician tasked with determining the logical relationship between two clinical statements based on a patient’s past medical history.\n"
4 "Analyze carefully whether the second statement (hypothesis) can be inferred from the first statement (premise).\n"
5 "Base your reasoning strictly on clinical knowledge, without making unwarranted assumptions.\n"

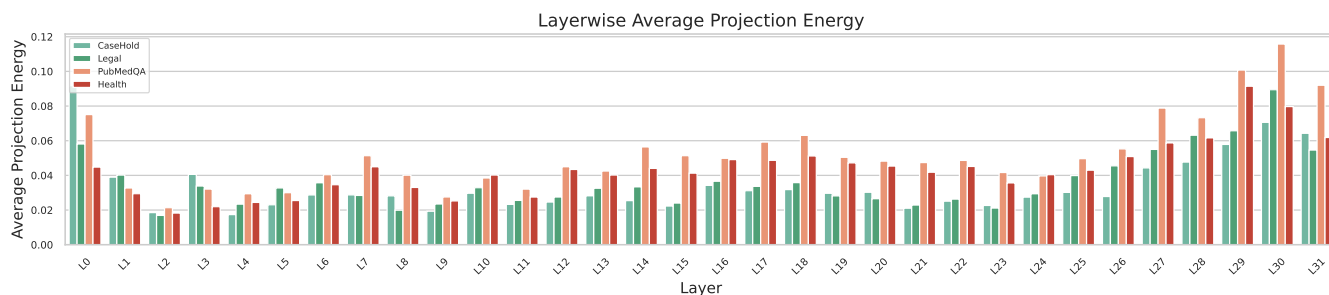


Figure 6: Projection energy across layers.

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6 "Your answer must be one of three
   options:\n"
7 "- 'entailment' if the hypothesis must
   be true given the premise,\n"
8 "- 'contradiction' if the hypothesis
   must be false given the premise,\n"
9 "- 'neutral' if the hypothesis could be
   true or false without certainty.\n"
10 "Fully consider all three possibilities
   and reason carefully before giving
   your final answer.\n\n"
11 "Input:\nPremise: '{sentence1}'\n
   nHypothesis: '{sentence2}'\n\n"
12 "Final Answer: Their relationship is {
   gold_label}"

```

HQS

```

1 "Below is an instruction that describes
   a task related to HealthCare, paired
   with further context. "
2 "Write a response that appropriately
   completes the request.\n\n"
3 "Instruction: Summarize the following
   consumer health question by
   generating a condensed version "
4 "that retains all critical information
   necessary to find correct and
   complete answers. "
5 "Focus on preserving key entities (e.g.,
   conditions, treatments, tests) and
   the main intent of the question,
   while omitting unnecessary peripheral
   details. "
6 "Write the summary fluently in natural
   language and avoid simply shortening
   without ensuring information
   completeness.\n\n"
7 "Please provide the shortened version
   directly.\n\n"
8 "Input: {input_question}\n\nShortened
   Question:"

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CaseHold

```

1 "Below is an instruction that describes
   a task related to making legal
   decisions based on a citing prompt. "
2 "Please read the citing prompt, and
   decide which holding statement best
   corresponds to it.\n\n"

```

```

3 "Instruction:\n"
4 "You are given a citing passage from a
   legal decision and five potential
   holding statements.\n"
5 "Your task is to carefully read the
   citing prompt and select the holding
   statement that best corresponds to it
   .\n"
6 "Focus on the precise legal principle,
   factual framing, and implications
   described.\n"
7 "Base your decision purely on the
   logical match between the citing
   prompt and the holding, avoiding
   irrelevant or merely similar content
   .\n\n"
8 "Your answer should be 0, 1, 2, 3, or
   4.\n\n"
9 "Input: citing_prompt: {citing_prompt},
   0: {holding_0}, 1: {holding_1}, 2: {
   holding_2}, 3: {holding_3}, 4: {
   holding_4}\n\n"
10 "Response: The answer is"

```

ContractNLI

```

1 "Below is an instruction that describes
   a task related to Contracts, paired
   with an input that provides context.
   "
2 "Write a detailed response that includes
   your reasoning process, then
   conclude with your final answer.\n\n"
3 "Instruction: Carefully analyze the
   relationship between the Contract
   Premise and Hypothesis. "
4 "Your final answer must be one of
   entailment, contradiction, or neutral
   . "
5 "Ensure that you fully consider all
   three relationships.\n\n"
6 "Input:\nPremise: '{sentence1}'\n
   nHypothesis: '{sentence2}'\n\n"
7 "Final Answer: Their relationship is "

```

BillSum

```

1 "Below is an instruction for summarizing
   a legislative bill.\n\n"
2 "Instruction:\n"

```

```

3 "Summarize the following legislative
   text by clearly identifying and
   explaining the *major actions,
   purposes, and effects* of the bill. "
4 "Focus on what the bill aims to achieve
   rather than technical details or
   administrative changes. "
5 "Paraphrase the information in a simple
   and accessible way as if writing for
   policymakers and the public. "
6 "Avoid copying long passages or citing
   subsection numbers unless necessary
   for understanding.\n\n"
7 "Input:\n"
8 "Bill Title: {title}\n\n"
9 "{input_text}\n\n"
10 "Response:"

```

- **ContractNLI:** The dataset is licensed under the CC BY 4.0 license.

Effect of Rank r on Partial Brain Surgeon

When the chosen rank r is substantially smaller than the number of pruned entries $|S_i|$, the resulting system becomes over-constrained, which can degrade downstream performance. To mitigate this, higher sparsity levels generally require a correspondingly larger rank r .

We evaluated the impact of varying r at 50% sparsity on Llama-3.2-1B. As shown in Figure 8, increasing the rank consistently improves relative performance. Each individual metric is recorded in table 11

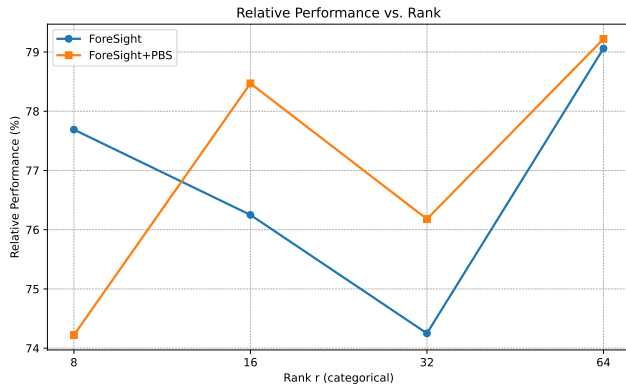


Figure 8: Relative performance of FourSight and FourSight + PBS at 50% sparsity on Llama-3.2-1B.

Discuss on the Artifacts

The license for the models and datasets used in this paper is as follows:

- **Llama-2-7b-hf:** The model is licensed under the LLaMA 2 community license.
- **Llama-3-8b:** The model is licensed under META LLaMA 3 community license.
- **PubMedQA:** The dataset is licensed under MIT license.
- **MedNLI:** The dataset is licensed under the Physionet Credentialed Health Data License Version 1.5.0.
- **HQS:** The dataset is licensed under Apache License 2.0.
- **CaseHold:** The dataset is licensed under Apache License 2.0.

Table 11: Performance metrics for different methods at various ranks r at 50% sparsity on Llama-3.2-1B.

Method	Rank	Med PPL	MedNLI Acc	MedNLI F1	PubMedQA Acc	PubMedQA F1	HQS R1	HQS R2	HQS RL	Rel%
Dense	8	10.288	0.5605	0.5464	0.6800	0.4714	0.2476	0.0896	0.2204	–
	16	10.145	0.5956	0.5759	0.6600	0.4674	0.2609	0.0852	0.2343	–
	32	10.029	0.6069	0.5986	0.6660	0.4656	0.2507	0.0809	0.2284	–
	64	9.902	0.5872	0.5733	0.6500	0.4575	0.2439	0.0755	0.2122	–
ours	8	14.716	0.4286	0.3961	0.5200	0.3710	0.2171	0.0688	0.1948	77.69
	16	14.706	0.4286	0.3961	0.5200	0.3710	0.2171	0.0688	0.1948	76.26
	32	14.685	0.4197	0.3620	0.5205	0.3716	0.2114	0.0672	0.1914	74.26
	64	14.834	0.3926	0.3921	0.5300	0.3972	0.2201	0.0693	0.1974	79.06
oursPBS	8	15.854	0.3643	0.3412	0.5720	0.4027	0.1960	0.0568	0.1757	74.22
	16	15.020	0.4365	0.4252	0.5395	0.3845	0.2206	0.0637	0.1980	78.47
	32	14.908	0.3968	0.3979	0.5505	0.3917	0.2117	0.0641	0.1894	76.18
	64	15.106	0.3954	0.3824	0.5490	0.3910	0.2227	0.0688	0.1998	79.22

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